# UNIVERSITY OF NAMIBIA

## FACULTY OF AGRICULTURE, ENGINEERING & NATURAL SCIENCES

## DEPARTMENT OF COMPUTING, MATHEMATICAL AND STATISTICAL SCIENCE


# TITLE OF THE THESIS

A thesis submitted in partial fulfilment of the requirements

for the degree of Bachelor of Science [Degree Name]


Student Name: [Full Name]

Student Number: [Student Number]

Supervisor: [Supervisor's Name]


Date: [Month, Year]

## DECLARATION

I, [Student Name], declare that this thesis is my original work and has not been submitted to any other institution for the award of a degree. All sources used in this study have been appropriately acknowledged.

Signature: _____ Date: _____

## ABSTRACT

[Write a concise summary of the study in approximately 200–300 words. Include the background, research problem, objectives, methodology, key findings, and main conclusion.]

Keywords: [Keyword 1]; [Keyword 2]; [Keyword 3]; [Keyword 4]

## ACKNOWLEDGEMENTS

[Acknowledge individuals and institutions that supported the research.]

## DEDICATION (OPTIONAL)

[Provide a dedication, if applicable.]

# TABLE OF CONTENTS

[Insert an automatically generated table of contents using Microsoft Word heading styles.]

## LIST OF FIGURES

[Insert the list of figures, if applicable.]

## LIST OF TABLES

[Insert the list of tables, if applicable.]

## LIST OF ABBREVIATIONS AND ACRONYMS

[List abbreviations and their meanings.]

# CHAPTER 1: INTRODUCTION

## 1.1 Introduction

Provide a brief overview of the research topic and introduce the context of the study.

## 1.2 Background of the Study

Explain the broader context of the problem, including relevant technological developments, industry trends, existing systems, or social and organisational challenges.

## 1.3 Problem Statement

Clearly describe the problem, who is affected, evidence that the problem exists, and why it needs to be addressed.

## 1.4 Aim of the Study

State the overall purpose of the research.

## 1.5 Research Objectives

List specific, measurable objectives that guide the study.

## 1.6 Research Questions

Develop questions that correspond to the research objectives.

## 1.7 Significance of the Study

Explain who will benefit from the research and how.

## 1.8 Scope of the Study

Define the geographical, technical, population, and functional boundaries of the study.

## 1.9 Limitations of the Study

Discuss challenges that may affect the research, such as time, resources, access to participants, or data availability.

# CHAPTER 2: LITERATURE REVIEW

## 2.1 Introduction

Introduce the purpose and structure of the literature review.

## 2.2 Theoretical Background

Discuss theories, models, or frameworks relevant to the research, where applicable.

## 2.3 Conceptual Framework

Present and explain relationships between important concepts or variables, where applicable.

## 2.4 Review of Existing Technologies

Discuss technologies, systems, tools, or platforms related to the research.

## 2.5 Review of Related Studies

Review previous studies, including their methods, findings, limitations, and relevance.

## 2.6 Review of Existing Systems or Applications

For development projects, examine similar systems, their features, strengths, weaknesses, and technologies.

## 2.7 Research Gap

Explain what is missing from existing research, systems, or knowledge.

## 2.8 Summary of the Chapter

Summarise the key findings from the literature review.

# CHAPTER 3: RESEARCH METHODOLOGY

## 3.1 Introduction

Introduce the selected methodology.

## 3.2 Research Approach

Explain whether the study uses quantitative, qualitative, mixed-methods, design science, or another appropriate approach.

## 3.3 Research Design

Describe the overall research design, such as a survey, case study, experiment, or design-and-development approach.

## 3.4 Target Population

Identify the population relevant to the study.

## 3.5 Sampling Techniques and Sample Size

Explain the sampling method, sample size, and relevant inclusion or exclusion criteria.

## 3.6 Data Collection Methods

Describe questionnaires, interviews, observations, document analysis, system logs, or other methods used.

## 3.7 Data Analysis Methods

Explain how the collected data will be analysed.

## 3.8 Software Development Methodology

For development projects, explain the selected methodology, such as Agile, Scrum, Waterfall, or Prototyping.

## 3.9 Ethical Considerations

Discuss consent, privacy, confidentiality, data protection, and research approval where applicable.

## 3.10 Chapter Summary

Summarise the chapter.

# CHAPTER 4: SYSTEM ANALYSIS AND DESIGN

## 4.1 Introduction

Introduce the system analysis and design activities.

## 4.2 Analysis of the Existing System

Discuss the current system or process, its problems, workflows, and weaknesses. (if there is one that exists)

### 4.2.1 Functional Requirements

Describe what the system must do.

### 4.2.2 Non-Functional Requirements

Describe quality attributes such as security, usability, performance, reliability, scalability, and maintainability.

## 4.3 System Design

Include relevant diagrams such as use-case, activity, class, sequence, data-flow, entity-relationship, or architecture diagrams.

## 4.4 Database Design

Where applicable, include the ER diagram, database schema, data dictionary, and table descriptions.

## 4.5 User Interface Design

Include wireframes, prototypes, or mock-ups and explain design decisions.

## 4.6 System Architecture

Explain the overall architecture of the proposed solution.

# CHAPTER 5: SYSTEM IMPLEMENTATION

## 5.1 Introduction

Introduce the implementation process.

## 5.2 Development Environment

Describe the programming languages, frameworks, databases, tools, operating systems, and version-control systems used.

## 5.3 System Implementation

Describe how the system was developed or configured.

## 5.4 Implementation of Major Features

Explain the development of key features and include appropriate screenshots or code excerpts.

## 5.5 Database Implementation

Describe the implementation of the database, where applicable.

## 5.6 Security Implementation

Explain authentication, authorisation, encryption, validation, or other security mechanisms.

## 5.7 Challenges Encountered During Development

Discuss technical challenges and how they were addressed.

# CHAPTER 6: SYSTEM TESTING AND EVALUATION

## 6.1 Introduction

Introduce the testing and evaluation activities.

## 6.2 Testing Strategy

Describe unit, integration, system, usability, performance, security, or user-acceptance testing as appropriate.

## 6.3 Test Cases

Document test cases using a table containing the test case ID, functionality, input, expected result, actual result, and status.

## 6.4 Testing Results

Present test results using tables, graphs, screenshots, or other suitable formats.

## 6.5 System Evaluation

Evaluate the system against the research objectives and requirements.

## 6.6 User Feedback and Usability Evaluation

Where applicable, report feedback from users who tested the system.

## 6.7 Discussion of Findings

Interpret the results and compare them with the literature review.

# CHAPTER 7: CONCLUSION AND RECOMMENDATIONS

## 7.1 Introduction

Introduce the final chapter.

## 7.2 Summary of the Study

Summarise the research problem, methodology, and major activities.

## 7.3 Summary of Key Findings

Present the major findings of the study.

## 7.4 Achievement of Research Objectives

Explain whether each objective was achieved, partially achieved, or not achieved, and provide evidence.

## 7.5 Conclusion

Provide a clear conclusion based on the findings.

## 7.6 Recommendations

Provide practical recommendations based on the research.

## 7.7 Suggestions for Future Research

Identify areas that future researchers could investigate.

## REFERENCES

List all sources cited in the thesis using APA 7th edition.

You should use credible academic sources and should not rely exclusively on websites or AI-generated references.

## APPENDICES

Include supporting materials that are too lengthy for the main body of the thesis.

- Appendix A: Research Questionnaire
- Appendix B: Interview Guide
- Appendix C: Additional System Screenshots
- Appendix D: **Selected** Source Code (Not all of it)
- Appendix E: Test Results
- Appendix F: Project User Manual

## RECOMMENDED FORMATTING GUIDELINES

| Element | Recommended Standard |
|---|---|
| Paper Size | A4 |
| Font | Times New Roman or Arial |
| Body Font Size | 12 pt |
| Line Spacing | 1.5 |
| Alignment | Justified |
| Margins | 2.54 cm |
| Chapter Headings | 16 pt, bold |
| Main Section Headings | 14 pt, bold |
| Subsection Headings | 12 pt, bold |
| Page Numbers | Bottom centre or bottom right |
| Referencing Style | APA 7th edition or departmental standard |
| Figure Captions | Below figures |
| Table Captions | Above tables |
| Language | Formal academic English |